

Fourth-Order Perturbations of Pure-Weyl Black Holes

Exact Ricci–Bach Composition, Horizon Pairing, and Endpoint Nonselection

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Abstract

We ask whether familiar one-ended Lorentzian endpoint conditions isolate the Einstein sector of four-dimensional pure Weyl gravity around a black hole. We first classify the Bach-flat locus in a declared static Laurent ansatz, recover the Mannheim–Kazanas component, and derive a residual-basic horizon generator. On its nondegenerate static domain, the associated Lee–Wald Hamiltonian, Wald entropy, and signed surface-gravity temperature obey an exact first law at every simple horizon. We correct an important sheet distinction: on the complete Laurent locus the metric is Einstein precisely when $\gamma = 0$ and $w = 1$, whereas $\gamma = 0$ alone suffices only on the Mannheim–Kazanas component through $w = 1$.

On Schwarzschild, linearizing the Bach tensor gives an exact Ricci-carrier composition. It defines an exact sequence from Bach solutions to curvature solutions, not a canonical Einstein/additional direct sum. In the axial $\ell = 2$ carrier, $\psi_{ab} = \delta R_{ab}$ obeys a second-order Lichnerowicz-type equation. For nonzero real frequency its horizon system has a two-dimensional analytic ingoing ψ -space, so future-horizon regularity does not force $\psi = 0$. The action-derived Lee–Wald current makes the Einstein self-block isotropic for conjugate Regge–Wheeler waves. Controlled finite-series fixtures at three frequencies exhibit nonzero Einstein–additional cross pairing and nonzero additional-sector pairing; these fixture results are not promoted to a symbolic all-frequency theorem.

At infinity the certified characteristic polynomial is repeated. Until its Jordan structure, metric reconstruction, finite-flux falloff, and polar analogue are completed, we claim only endpoint nonselection: horizon analyticity together with the presently tested real-frequency leading asymptotics does not imply $\delta R_{ab} = 0$. This is not a no-go against all local causal truncations; the local Cauchy restriction $\psi|_{\Sigma} = \nabla_n \psi|_{\Sigma} = 0$ selects the Einstein kernel at the linear level. We make no claim about stability, quasinormal spectra, nonlinear closure, Hawking radiation, or quantum unitarity.

AI authorship and computational accountability

The named AI systems developed the programme, derivations, symbolic implementations, independent verification architecture, claim audits, and this manuscript. Asger Alstrup Palm commissioned and coordinated the work and is the corresponding human contact. Every certified statement below is bound to a machine-readable certificate and a separately implemented verifier. The cross-flux result uses controlled high-precision series evaluation with exact recurrence data and independent frequencies; it is reported as fixture-level evidence, not interval-certified nonvanishing. All local

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tensor identities and polynomial classifications are exact. The manuscript remains conditional on final human review.

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1 The physical question

Fourth-order gravitational equations contain more local solutions than the Einstein equations. A common proposal is to retain the Einstein branch by a boundary condition. Around a one-ended Lorentzian black hole, a first test is:

Do future-horizon analyticity and the same leading outer admissibility tests used for Einstein waves force the additional Ricci carrier to vanish?

The answer in the certified Schwarzschild axial $\ell = 2$ mode laboratory is no: the curvature carrier has regular ingoing horizon data, the leading real-frequency symbol does not separate it from the Einstein carrier, and controlled fixtures have nonzero action-derived horizon pairing. This is an endpoint-admissibility result, not a classification of all local differential boundary conditions. In particular, setting the Ricci carrier and its normal derivative to zero on a Cauchy surface is a

local linear Einstein-sector restriction. Whether that restriction is natural, interaction-stable, or compatible with a physical asymptotic phase space is open.

1.1 Claims and dependency tags

The static family and local operator identities carry LOCAL-ALGEBRAIC. Harmonic branch, horizon, current, and asymptotic statements also carry REDUCED-MODE. No result in this manuscript is promoted to LORENTZIAN-CAUSAL: there is no complete exterior Green-hyperbolicity theorem, metric-falloff phase space, or classified set of local differential boundary operators.

2 Relation to existing branch-selection programmes

The static Bach-flat family itself is classical [2, 3]. Conformal-gravity thermodynamics with additional spin-two hair has also been studied in asymptotically AdS and Einstein–Weyl settings [14]. Our static contribution is therefore not a new family: it is the exact residual-basic normalization and simultaneous signed first-law identity within a declared local quotient. A physical asymptotic normalization has not yet been selected. In particular, Lü, Pang, Pope, and Vázquez-Poritz organize AdS thermodynamics with an independent massive-spin-two-hair conjugate pair. Our one-term identity instead follows after quotient-basic normalization on a restricted static parameter family. The different boundary ensemble, generator normalization, and independent variables mean that the two first laws are complementary, not contradictory.

Maldacena’s Einstein-from-conformal construction selects an Einstein sector by Euclidean AdS or late-time de Sitter boundary data [8]. Critical- and logarithmic-gravity analyses similarly separate massless, massive, and generalized modes by asymptotic conditions [9, 13, 10, 11, 12]. The null Einstein self-pairing and nonzero cross pairing found below should be read in that established degenerate-operator context, rather than as the first appearance of null massless modes paired with generalized solutions. Our different question is whether one-ended Lorentzian Schwarzschild endpoint tests force the Ricci carrier to vanish.

Using δR_{ab} to expose higher-derivative spin-two modes is also a known strategy in fourth-order gravity [15]. Previous Weyl-black-hole quasinormal analyses often deliberately obtained a second-order master equation by conformal transport rather than retaining the full fourth-order Bach solution space [16]. Effective-Einstein perturbations of conformally related nonsingular metrics [17] and scalar or electromagnetic test-field spectra [18] are different problems: test-field stability is not stability of the dynamical fourth-order Bach sector. Contemporary quadratic-gravity ringdown work finds both ordinary and additional spin-two sectors [19], while time-domain quadratic-gravity studies track massive-sector tails [20]. The prospective advance here is narrower: the exact pure-Weyl Ricci–Bach sequence, its action-derived Schwarzschild horizon pairing, and failure of the tested endpoint conditions to imply $\delta R_{ab} = 0$. The first-order-system analysis follows the general modified-gravity asymptotic methodology of [21]; in particular, that methodology makes the unresolved repeated-root Jordan audit an essential next step rather than a technical detail. Precise decay of the metric coefficients must ultimately be compared with the Schwarzschild Einstein benchmark [22], not inferred from bounded curvature carrier variables alone.

3 Pure Weyl gravity and its covariant current

We use signature $(-+++)$ and action

$$S_W[g] = \alpha \int_M \sqrt{-g} C_{abcd} C^{abcd} d^4x. \quad (1)$$

The field equation is the Bach equation $B_{ab} = 0$, with convention

$$B_{ab} = \nabla^c \nabla^d C_{acbd} + \frac{1}{2} R^{cd} C_{acbd}. \quad (2)$$

The covariant presymplectic current is obtained from the literal variation of the curvature-momentum potential of this action. Its normalization is fixed by the off-shell Green identity

$$\partial_t F^t(h_A, h_B) + \partial_r F^r(h_A, h_B) = 4\alpha \int_{S^2} \sqrt{q} (h_B^{ab} \delta B_{ab}[h_A] - h_A^{ab} \delta B_{ab}[h_B]). \quad (3)$$

Thus the current is conserved for pairs of linearized Bach solutions and is not a freely rescaled master-equation Wronskian.

4 Static spherical Bach-flat family

In Schwarzschild gauge,

$$ds^2 = -B(r)dt^2 + B(r)^{-1}dr^2 + r^2 d\Omega_2^2, \quad (4)$$

the declared Laurent ansatz is

$$B(r) = w - \frac{u}{r} + \gamma r - kr^2 + \frac{c_2}{r^2} + c_3 r^3. \quad (5)$$

Theorem 4.1 (Complete Bach-flat locus in the Laurent ansatz). *The metric is Bach-flat if and only if*

$$c_2 = c_3 = 0, \quad w^2 + 3u\gamma = 1. \quad (6)$$

The Mannheim–Kazanas parametrization is recovered by

$$w = 1 - 3\beta\gamma, \quad u = \beta(2 - 3\beta\gamma).$$

On the complete Laurent locus the metric is Einstein if and only if

$$\gamma = 0, \quad w = 1. \quad (7)$$

On the Mannheim–Kazanas component through $w = 1$, this reduces to the single condition $\gamma = 0$.

Completeness here refers only to the Laurent ansatz. The certificate does not claim a new proof of the global static spherical classification. The sheet qualification in (7) follows already from the scalar curvature

$$R = 12k - \frac{6\gamma}{r} + \frac{2(1-w)}{r^2}. \quad (8)$$

In particular, the allowed point $(\gamma, w) = (0, -1)$ is not Einstein. The original static certificate’s phrase “Einstein iff $\gamma = 0$ ” is therefore used only on its explicitly parametrized Mannheim–Kazanas sheet; the manuscript does not propagate it to the complementary Laurent sheet.

The form-preserving local $\text{Diff} \times \text{Weyl}$ transformations act as dilation and translation on the cubic

$$Q(x) = -ux^3 + wx^2 + \gamma x - k, \quad x = r^{-1}.$$

Their orbit rank is two. A single continuous invariant may be taken as

$$J = u^2 \text{disc}(Q),$$

while $\text{disc}(Q) = 0$ is the degenerate-horizon locus. The conformal translation may fail globally where its Weyl factor vanishes; it is used only as a local orbit statement.

An exact non-Einstein three-horizon fixture is

$$(\beta, \gamma, k) = \left(\frac{3}{2}, \frac{12}{19}, \frac{1}{19} \right), \quad rB(r) = -\frac{1}{19}(r-1)(r-3)(r-8).$$

All curvature invariants are finite at the simple horizons. In ingoing Eddington–Finkelstein coordinates,

$$ds^2 = -B(r)dv^2 + 2dvdr + r^2 d\Omega_2^2,$$

the metric and Bach equation are regular there.

5 Residual-basic energy and the static first law

For the chart generator ∂_t , the bare static Lee–Wald one-form is not integrable. The obstruction disappears when the generator is normalized consistently with the residual quotient.

Theorem 5.1 (Normalized static generator). *On a connected component with $u \neq 0$, choose the oriented generator*

$$\chi = u \partial_t, \quad u = \beta(2 - 3\beta\gamma). \quad (9)$$

Residual basicness and dilation weight determine the general normalization as $uf(J)\partial_t$. We select $f(J) = 1$. Its Hamiltonian is

$$H = -16\pi\alpha\beta^2 D_2, \quad D_2 = 9\beta^2\gamma^2 k - \beta\gamma^3 - 12\beta\gamma k + \gamma^2 + 4k. \quad (10)$$

At every simple horizon r_h , the Wald entropy

$$S_h = 64\pi^2\alpha\beta \frac{2 - 3\beta\gamma + \gamma r_h}{r_h} \quad (11)$$

and normalized temperature

$$T_h = \frac{uB'(r_h)}{4\pi} \quad (12)$$

satisfy

$$dH = T_h dS_h \quad (13)$$

identically modulo $B(r_h) = 0$.

Here T_h is the *signed* surface-gravity temperature appropriate to the oriented first-law identity; the positive Hawking temperature would use $|uB'(r_h)|/(4\pi)$. On a component where $u < 0$, future-directedness uses $-\chi$, flipping both H and T_h and preserving the first law. The locus $u = 0$ is excluded because this normalization degenerates. For another allowed $f(J)$, one has $T_f = f(J)T_h$ and a reparametrized Hamiltonian satisfying $dH_f = f(J)dH$; the first law remains true. Thus

residual basicness fixes a class, not a unique physical normalization in the absence of an asymptotic clock.

This is a theorem on the static parameter slice, not yet a general physical-process first law. Its linear dynamical extension nevertheless passes a strong gauge audit: arbitrary time-dependent spherical Weyl rescalings and diffeomorphisms have exactly zero charge and flux, the entropy is conformally invariant on the symbolic family, and the normalized generator extension is unique at linear order under the declared boundary scalar rule.

6 Ricci–Bach composition and the axial exact sequence

The parity-specific formulas are restrictions of one Ricci-flat identity.

Proposition 6.1 (Linearized Bach tensor on a Ricci-flat background). *For a perturbation of a four-dimensional Ricci-flat metric,*

$$\delta B_{ab} = \frac{1}{2}\square(\delta R_{ab}) + C_{acbd}(\delta R)^{cd} - \frac{1}{6}\nabla_a\nabla_b\delta R - \frac{1}{12}g_{ab}\square\delta R. \quad (14)$$

Proof. Rewrite the four-dimensional Bach tensor in Ricci variables using the contracted Bianchi identity, then retain the terms linear in Ricci curvature about $R_{ab} = 0$. Background derivatives and the curvature commutator give the displayed Weyl action. The repository verifies (14) independently, component by component, on the complete axial and polar $\ell = 2$ Regge–Wheeler-gauge carriers. That computer check is carrier-complete but is not presented as a machine proof for arbitrary harmonic content. $\square$

We now specialize the background to Schwarzschild and use axial Regge–Wheeler gauge,

$$h_{t\phi} = h_0(t, r)S_{\ell m}, \quad h_{r\phi} = h_1(t, r)S_{\ell m}.$$

The $\ell = 2$ linearized Bach tensor is fourth order, trace-free, and obeys the exact linearized Noether identity. Eliminating the constraint on the Einstein subbranch reproduces the Regge–Wheeler equation for $\Psi = Bh_1/r$,

$$B(B\Psi)' + (\omega^2 - V_{\text{RW}})\Psi = 0, \quad V_{\text{RW}} = B\left(\frac{6}{r^2} - \frac{6m}{r^3}\right). \quad (15)$$

Corollary 6.2 (Axial composition). *For the certified axial carrier, parity gives $\delta R = 0$, so*

$$\boxed{\delta B_{ab} = \frac{1}{2}\square(\delta R_{ab}) + C_{acbd}(\delta R)^{cd}.} \quad (16)$$

Consequently the Einstein kernel $\delta R_{ab} = 0$ injects into the Bach kernel, while its curvature image obeys

$$\frac{1}{2}\square\psi_{ab} + C_{acbd}\psi^{cd} = 0, \quad \psi_{ab} := \delta R_{ab}. \quad (17)$$

Let $\mathcal{C}_{\text{ax}}$ be the fixed axial carrier and define

$$\mathcal{H}_B = \{h \in \mathcal{C}_{\text{ax}} : \delta B[h] = 0\}, \quad \mathcal{H}_E = \{h \in \mathcal{H}_B : \delta R[h] = 0\}, \quad (18)$$

$$\mathcal{H}_R = \ker\left(\frac{1}{2}\square + C\circ\right) \cap \text{im}(\delta R|_{\mathcal{C}_{\text{ax}}}). \quad (19)$$

Then the invariant statement supplied by the composition is the exact sequence

$$0 \longrightarrow \mathcal{H}_E \longrightarrow \mathcal{H}_B \xrightarrow{\delta R} \mathcal{H}_R \longrightarrow 0. \quad (20)$$

It does *not* supply a canonical direct sum. A metric lift h_X of $\psi \in \mathcal{H}_R$, when it exists in the chosen domain, may be shifted by any $h_E \in \mathcal{H}_E$. Global lift existence, boundary conditions, and a gauge-independent splitting must therefore be proved separately. In particular, “additional branch” below refers either to the curvature image $\mathcal{H}_R$ or to an explicitly constructed lift, never to an assumed canonical complement.

The identity is componentwise and off shell in the perturbation. It does not extend in this two-term form to the certified non-Einstein static fixture. Thus the clean composition is a Ricci-flat-background theorem.

7 The Ricci carrier reaches the future horizon

In ingoing coordinates (v, r) , Fourier modes are written $e^{i\omega v}$. The extra carrier’s divergence constraint solves its third component polynomially; the apparent Schwarzschild-coordinate pole is a chart artifact.

Theorem 7.1 (Analytic ingoing curvature family). *For axial $\ell = 2$ and $\omega \in \mathbb{R} \setminus \{0\}$, the future horizon $r = 2m$ is a regular singular point of the extra first-order radial system. Its residue spectrum is*

$$\{0, 0, -4im\omega, -2 - 4im\omega\}, \quad (21)$$

and the zero eigenspace has dimension two. Hence there is a two-parameter family of ψ -solutions analytic at the future horizon, with no leading logarithm at the zero exponent.

The restriction excludes $\omega = 0$, where the algebraic multiplicity of the zero residue exponent changes, and makes no claim at resonant complex frequencies. Most importantly, the theorem counts the curvature carrier, not a quotient of metric solutions. The currently certified dimensions are:

Space	Ingoing/analytic dimension	Status
Einstein Regge–Wheeler carrier	1	second-order control
Ricci carrier ψ	2	exact residue calculation
Full Bach metric solution space	not certified	reconstruction/domain open
Metric quotient by the Einstein kernel	not certified	lift and gauge open

Thus future-horizon regularity does not force $\psi = 0$. Explicit metric lifts are constructed in the flux fixtures below, but the exact sequence (20) has not yet been split globally.

8 Action-derived horizon flux

Substituting two on-shell Regge–Wheeler solutions into the current (3) gives

$$F_{EE}^r = -\frac{192\pi\alpha(\omega_1^2 - \omega_2^2)}{5\omega_1\omega_2 r} \Psi_1 \Psi_2. \quad (22)$$

For conjugate pairs $\omega_2 = \pm\omega_1$, this vanishes identically.

Theorem 8.1 (Exact Einstein isotropy). *On Schwarzschild in axial $\ell = 2$, the pure-Weyl Lee–Wald pairing of an Einstein Regge–Wheeler mode with its conjugate vanishes exactly:*

$$F^r(E_\omega, \overline{E_\omega}) = 0. \quad (23)$$

Thus the Einstein line is isotropic in the presymplectic structure inherited from the pure C^2 action. This statement does not identify the Einstein symplectic form with the ordinary Einstein–Hilbert one.

Proposition 8.2 (Controlled horizon-pairing fixtures). *For $m = 1$, axial $\ell = 2$, and conjugate-frequency Hermitian pairings, order-16 ingoing series give*

$$F^r(E_\omega, \overline{X_\omega}) \neq 0, \quad F^r(X_\omega, \overline{X_\omega}) \neq 0 \quad (24)$$

at producer frequencies $\omega = 3/5, 2/7$; an independent verifier repeats the nonvanishing gates at $\omega = 1/2$. At the two stored frequencies the values, divided by $\pi\alpha$ and sampled at two interior radii, are approximately

ω	$F^r(X, \bar{X})/(\pi\alpha)$	$F^r(E, \bar{X})/(\pi\alpha)$
3/5	$-330.58i, -329.98i$	$-218.17 + 43.55i, -215.97 + 51.53i$
2/7	$-71.92i, -71.95i$	$-162.82 + 27.99i, -163.92 + 33.03i$

The exact Regge–Wheeler null identity is the in-run control. These data are controlled numerical fixtures: they prove neither symbolic nonvanishing for all real ω nor an interval-certified error bound.

For a chosen lift basis (E, X) , the horizon block has the form

$$\Omega_h = \begin{pmatrix} 0 & a \\ -\bar{a} & b \end{pmatrix}. \quad (25)$$

Under the admissible lift change $X \mapsto X + cE$, the cross entry a is invariant while b changes. The robust fixture-level conclusions are therefore Einstein isotropy, nonzero cross pairing, and full rank of the tested two-dimensional block. The sign of iF_{XX}^r for one chosen lift is basis dependent and is not promoted to a branch signature or Hilbert norm.

The stored report also records a repaired calculation: an earlier flux evaluation dropped the Fourier factor $e^{i\omega t}$, causing all time derivatives to vanish silently. The exact Regge–Wheeler null control exposed the defect; all displayed fixtures use the repaired full-time current.

9 Asymptotic symbol and endpoint nonselection

At real frequency and large radius, the extra carrier has dispersion

$$(\lambda_r^2 - \omega^2)^2 = 0 \quad (26)$$

and growth exponents

$$\sigma \in \{\pm 2im\omega, \pm 2im\omega - 1\}. \quad (27)$$

The simple-root reading would identify Coulomb logarithmic phases with amplitudes of order r^0 and r^{-1} . But the characteristic is *double*. The polynomial alone does not determine geometric multiplicity. As in $(D^2 + \omega^2)^2 h = 0$, a Jordan chain may produce $r_* e^{\pm i\omega r_*}$ even though every characteristic exponent is oscillatory. This question is now certified: the integer-spaced exponent resonances within each characteristic sector are consistent, so the *carrier* formal fundamental systems at infinity are log-free in both parity sectors (axial at symbolic real frequency; the polar $\mu = -2\omega$ sector at two rational fixtures) — no Jordan blocks at the formal carrier level.

Theorem 9.1 (Endpoint regularity-and-leading-symbol nonselection). *On the declared Schwarzschild axial $\ell = 2$, nonzero-real-frequency mode carrier, future-horizon analyticity does not imply $\delta R_{ab} = 0$. Moreover, the presently certified leading characteristic polynomial and formal exponents do not supply an outer condition that distinguishes the Ricci carrier from the Einstein characteristic. Hence these tested local endpoint diagnostics do not select the Einstein kernel.*

This deliberately does *not* say that every ordinary asymptotically flat radiation condition admits an additional metric mode. That stronger statement requires: (i) the full asymptotic Jordan decomposition; (ii) reconstruction of h from $\psi = \delta R[h]$; (iii) falloff in a regular tetrad or asymptotic gauge; and (iv) finite Lee–Wald flux at null infinity. Three of these four stations are now certified, and they *reverse* the expectation the nonselection theorem might have suggested. (i) The carrier Jordan decomposition is log-free in both parities. (ii) Reconstruction is not: solving $\delta R[h] = \psi$ at infinity, the pure-power ansatz is inconsistent and the single-log ansatz succeeds with nonzero log part in both characteristic sectors — the composed metric carries *logarithmic tails*, the inhomogeneous realization of the double characteristic root. (iv) Substituting the formal pair classes into the sphere-integrated slice density F^v (axial $\ell = 2$, $\omega = 3/5$ fixture): the Einstein self-class falls as r^{-2} in both sectors and has *finite* slice norm, while every class involving the composed modes has leading behavior between $r^0 \ln r$ and r^2 and *diverges*, invariantly under Einstein-shifts of the composed representative.

Theorem 9.2 (Fixture norm-selection at infinity). *On the declared axial $\ell = 2$, $\omega = 3/5$ mode laboratory, the finite-slice-norm asymptotic class of the sphere-integrated presymplectic density contains exactly the Einstein sector.*

The *finite* side of this selection is no longer fixture-bound. Driving the certified sphere-integrated Lee–Wald slice density F^v with the certified Einstein profiles at *symbolic* real ω , retaining all boundary and derivative-of-curvature terms (no leading-exponent shortcut), the axial Einstein conjugate pairs fall as

$$F^v[E_0 \times E_0] \sim r^{-2}, \quad F^v[E_2 \times E_2] \sim r^{-2}, \quad (28)$$

with an ω -independent integer leading power $-2 < -1$ and a nonzero, real-pole-free leading coefficient. Because ω enters the extra-branch carrier exponents $\pm i\omega$, $\pm 2i\omega$ only through the imaginary tortoise phase, the amplitude real parts are ω -independent: the Einstein slice norm is *finite for every real $\omega \neq 0$* , with $\omega = 0$ the excluded exceptional carrier.

Theorem 9.3 (Axial symbolic-frequency Einstein finiteness). *On the declared axial $\ell = 2$ sector at symbolic real $\omega \neq 0$, the literal Lee–Wald Einstein slice norm is finite ($F^v \sim r^{-2}$), ω -independently. The complementary divergent classes and the polar sector remain at the $\omega = 3/5$ fixture level.*

The Einstein–extra symplectic pairing carries an invariant of the same data. On $\text{span}(E, X)$ the Lee–Wald Gram is $G = \begin{pmatrix} 0 & a \\ a & b \end{pmatrix}$, with $a = a(\omega) = iF^r(E, X)/(\pi\alpha)$ the exact rational cross invariant (nonzero for every real $\omega \neq 0$) and b the representative-dependent extra self-pairing. Then $\det G = -|a|^2 < 0$ strictly, so G has *rank two and signature* $(1, 1)$, *independent of the representative b* . The rank and signature are invariant; the sign of the extra self-pairing b is *not*, and no such basis-dependent sign is claimed.

Thus the two ends of the exterior play opposite roles: the tested horizon and leading-symbol diagnostics do not select the Einstein kernel, but symplectic-norm finiteness at infinity does — as a phase-space normalization, not a local differential boundary condition — and on the finite-flux phase space the additional solution, admitted at the horizon, is excluded at infinity by norm finiteness rather than by any local boundary condition. Station (iii), falloff in a regular tetrad or

asymptotic gauge, remains open; the divergent-class table and the polar recomputation remain at the fixture level; and the exact global horizon-to-infinity connection map (a confluent-Heun connection) remains the open transcendental object. Nor is there a no-go against local causal selection: it is now a certified statement. On the globally hyperbolic exterior with Cauchy surface $\Sigma = \{t = \text{const}, r > 2m\}$, the local Cauchy condition

$$\psi|_{\Sigma} = 0, \quad \nabla_n \psi|_{\Sigma} = 0 \quad (29)$$

propagates $\psi = 0$ in the axial sector unconditionally: the carrier operator equals $\frac{1}{2}\square + C \circ (-)$ exactly there (the trace vanishes identically on the class), is normally hyperbolic, and its Bianchi constraint obeys the exact transport identity $\nabla^a(L\psi)_{a\varphi} = \frac{1}{2}\square B_{\varphi}$. In the polar sector the operator is identically traceless at the PDE level, the conformal direction is uncontrolled, and $\psi_{\text{conf}}(t^4\chi(r)P_2)$ is an exact nonuniqueness witness with vanishing Cauchy data; the certified conformal trace relation $S(\psi_{\text{conf}}(\Phi)) = -3\square\Phi$ then reduces any zero-data solution to the conformal-gauge orbit, so local Cauchy data selects the Einstein image exactly *modulo conformal gauge* (and exactly on the traceless slice). Dropping the normal-derivative datum is certified to fail by a time-odd mode witness. The open physical questions are whether this restriction is preserved by nonlinear interactions and whether it follows from a differentiable asymptotic phase space rather than being imposed by hand.

10 Polar chain: composition, reach, current, and disposition

For polar $\ell = 2$ Regge–Wheeler-gauge perturbations, the trace variation δR need not vanish. The axial formula acquires two universal trace terms.

Corollary 10.1 (Polar trace-coupled composition). *On the certified polar $\ell = 2$ carrier, Proposition 6.1 gives*

$$\delta B_{ab} = \frac{1}{2}\square(\delta R_{ab}) + C_{acbd}(\delta R)^{cd} - \frac{1}{6}\nabla_a\nabla_b\delta R - \frac{1}{12}g_{ab}\square\delta R. \quad (30)$$

The Einstein polar kernel injects by $\delta R_{ab} = 0$, while the curvature image obeys a second-order trace-coupled Lichnerowicz system for $(\delta R_{ab}, \delta R)$.

The polar chain is now certified through the same stations as the axial chain. Four structural facts organise it.

(i) *Polar Einstein kernel.* In the t -chart Regge–Wheeler polar gauge the traceless angular row of δR_{ab} equals $(H_0 - H_2)/2$, forcing $H_2 = H_0$; the remaining rows reduce exactly to a two-dimensional first-order system $dY/dr = M(r)Y$, $Y = (K, H_1)$, with H_0 algebraic. In horizon-adapted variables (K, BH_1) the ingoing residue spectrum is $\{0, -4im\omega\}$, matching the axial Regge–Wheeler control. A Schrödinger-form master scalar with ω -free potential was *not* found within the searched bounded rational ansatz classes; that anchor is recorded fail-closed open and nothing below depends on it.

(ii) *Conformal gauge and polar reach.* The polar carrier system on the Bianchi-constrained $(\delta R_{ab}, \delta R)$ is exactly three second-order equations in four radial functions: the operator (30) is identically traceless and divergence-free on the constraint surface, and the one-function underdeterminacy is linearized conformal gauge ($h = \Phi g$ is pure gauge and $\psi_{\text{conf}} = \delta\text{Ric}[\Phi g]$ annihilates every row for arbitrary Φ); this direction is absent in the axial sector. On the algebraic traceless slice the principal part diagonalises and the future horizon is a regular singular point with residue spectrum $\{0, 0, 0, 1 - 4im\omega, -1 - 4im\omega, -3 - 4im\omega\}$. Quotienting the regular conformal-gauge direction leaves a *two*-parameter physical ingoing-analytic polar carrier family at every $\omega \in \mathbb{R} \setminus \{0\}$ — the polar twin of the axial reach theorem.

(iii) *Polar current.* The sphere-integrated action-derived polar bilinear obeys the same off-shell 4α identity. On shell of the two-dimensional polar Einstein system every bilinear coefficient carries the factor $(\omega_1 + \omega_2)$: the polar Einstein self-block is exactly null for conjugate pairs, as in the axial sector. In addition the conformal direction Φg pairs to zero with an *arbitrary* off-shell polar perturbation: the sphere-integrated polar presymplectic form needs no conformal quotient at the bilinear level. Composition fixtures ($\omega = 3/5$, two interior radii, exact arithmetic) certify that the realized Ricci image closes on the full three-dimensional analytic carrier space — every ingoing-analytic polar carrier mode lifts to a metric perturbation with all seven δRic rows verified — and that the Einstein–carrier cross pairing is nonzero. Cross-pairing nonvanishing is representative-independent because the Einstein block is null; the positive carrier self-norms are certified at the canonical composed representatives only, and the representative-invariant sign theory is open.

(iv) *Polar disposition.* The polar carrier’s asymptotic symbol is $(\lambda^2 - \omega^2)^3$: it propagates on the Einstein characteristics, with Coulomb log-phases and amplitude falloffs r^{-1}, r^{-2}, r^{-3} — all decaying at real frequencies, strictly stronger than the axial r^0, r^{-1} case. The tested endpoint diagnostics (future-horizon analyticity, leading outer symbol and falloff class) therefore fail to separate the polar Ricci carrier from the polar Einstein carrier, exactly as in the axial sector: the endpoint-nonselection result is now *parity complete* at the $\ell = 2$ linear mode level. As before, this is not a classification of all local differential boundary operators.

(v) *Polar asymptotics and norm selection.* The asymptotic gate now closes in the polar sector as well ($\omega = 3/5$ fixture). Solving $\delta\text{Ric}[h] = \psi$ at infinity for each leading carrier formal solution, the polar composed lift is *power-enhanced*: in the $\mu = 0$ sector the pure-power and logarithmic ansätze at the naive base all fail, and the lift requires exactly one extra power of r together with a single logarithm — the inhomogeneous realization of the certified rank-one resonance (“at most one power of enhancement”) — while the $\mu = -2\omega$ sector lifts as a pure oscillatory power with neither enhancement nor logarithm, in contrast to the axial sector’s two-sided logarithmic tails. The exact conformal-gauge lift ($h = \Phi g, \square\Phi = 0$) passes through the same machinery as a pure power, an internal control that also guards the source-series depth (the derived carrier sources carry positive r -weights, so shallow carrier jets corrupt the resonant staircase; the certified computation uses depth-12 jets with recorded validity margins). Substituting the conjugate pair classes into the sphere-integrated slice density F^v : the $\mu = 0$ Einstein self-pair is *identically zero* in the slice density — an extra degeneracy of that sector — and the $\mu = -2\omega$ Einstein self-pair falls as r^{-2} , exactly the axial Einstein behavior, so the Einstein slice norm is finite in both sectors (the certified vanishing of the polar Einstein-branch *radial* flux for conjugate pairs is a separate exact statement); meanwhile every pair involving a composed mode diverges, with leading powers r^1 to r^4 , over all carrier representatives.

Theorem 10.2 (Polar fixture norm-selection at infinity). *On the declared polar $\ell = 2$, $\omega = 3/5$ mode laboratory, the finite-slice-norm asymptotic class of the sphere-integrated presymplectic density contains exactly the Einstein sector: the norm-selection of the axial fixture theorem is parity complete at the fixture level.*

11 Horizon monodromy temperature

One statement of Hawking type is available at the REDUCED-MODE level without any Lorentzian quantum construction. The surface gravity of the certified background is $\kappa = 1/(4m)$ with $T_H = \kappa/2\pi = 1/(8\pi m)$; the geometric-clock value agrees exactly with the certified normalized-frame first-law temperature.

Theorem 11.1 (Monodromy universality). *Under the horizon continuation $\rho \rightarrow e^{2\pi i}\rho$, every certified ingoing residue exponent of the four $\ell = 2$ mode families (axial and polar, Einstein and Ricci carrier) has continuation factor exactly 1 or $e^{8\pi m\omega} = e^{\omega/T_H}$, and every family contains exponents with the nontrivial factor.*

The integer parts of the exponents are monodromy-trivial; the universal $-4im\omega$ part carries the thermal factor. Hence the Boltzmann ratio $|\beta/\alpha|^2 = e^{-\omega/T_H}$ of the Damour–Ruffini continuation is identical for the Einstein and additional curvature families in both parity sectors: at the mode level, the additional carrier is thermally weighted at exactly the Hawking temperature, and by the certified nonzero mixed pairings it participates in the horizon flux with that weight. This is a mode-level (REDUCED-MODE) statement only: no Hadamard state, renormalized stress tensor, grey-body factor, luminosity, back-reaction, or LORENTZIAN-CAUSAL Hawking theorem is claimed, and none exists in the repository.

12 Result ledger

Question	Certified answer	Scope
Static Bach-flat black holes?	exact Laurent-family classification and three-horizon fixture; Einstein sheet corrected	local/static
Differentiable energy?	residual-basic normalized generator, exact static first law	$u \neq 0$; signed orientation; $f(J)$ freedom
Einstein axial waves present?	yes; exact Regge–Wheeler injection	Schwarzschild, axial $\ell = 2$
Additional axial content?	nonzero second-order Ricci carrier; no canonical metric split	same
Reach future horizon?	two analytic ingoing ψ amplitudes	nonzero-real-frequency carrier
Carry current?	Einstein self-block exactly null; mixed/additional blocks nonzero in fixtures	exact identity plus three frequencies
Selected by tested endpoints?	no: horizon and leading-symbol tests do not force $\psi = 0$	not a general causal no-go
Asymptotic metric radiation?	carrier formal systems log-free; composed metric has logarithmic tails (axial) or one power of enhancement with a single log (polar $\mu = 0$); finite-slice-norm class at infinity is exactly the Einstein sector in both parities, the polar $\mu = 0$ Einstein pair identically zero in the slice density	axial Einstein finite side symbolic ($F^v \sim r^{-2}$, all real $\omega \neq 0$); divergent classes and polar at fixture $\omega = 3/5$; tetrad falloff and phase-space construction open
Einstein–extra pairing invariant?	Lee–Wald Gram rank two, signature $(1, 1)$, representative-independent; extra self-pairing sign not invariant	axial $\ell = 2$, real $\omega \neq 0$
Complex-frequency structure?	axial modes and cross current continue meromorphically, exact singular set $\{i, i/2\}$ (polar not activated); Einstein BVP well-posed modulo discrete Wronskian zeros, additional branch log-tail obstructed	REDUCED-MODE; not a spectrum
Static ($\omega = 0$) sectors?	log-classified: two-parameter log-free families both parities; polar Einstein variables degenerate	classification only; static composition open
Polar composition?	exact trace-coupled identity; realized image closes on the full analytic carrier space	Schwarzschild, polar $\ell = 2$
Polar reach and current?	two physical ingoing amplitudes modulo conformal gauge; Einstein self-block null; nonzero mixed fixtures	conformal direction an exact off-shell degeneracy
Polar disposition?	Einstein characteristics, all-decaying Coulomb asymptotics; endpoint nonselection parity complete	real frequencies, $\ell = 2$
Thermal weighting?	monodromy factor $e^{-\omega/T_H}$ universal across branches and parities	REDUCED-MODE; no quantum state
Stable/ringing?	not established	complex frequencies open; no QNM

13 What is not proved

The paper does not establish:

- a complete exterior initial-boundary well-posedness theorem: the Einstein-branch complex-

frequency boundary-value problem is well-posed only *modulo* the discrete zeros of the connection Wronskian (the confluent-Heun connection, uncomputed), and the additional-branch outgoing condition is certified *ill-defined* by the log tails rather than resolved;

- general ℓ, m , the $\omega = 0$ sector, or a non-Einstein background Ricci-carrier composition;
- a canonical splitting of the Bach solution space, or complete metric reconstruction from every Ricci-carrier solution;
- a representative-invariant sign theory for the carrier self-pairing (the null-quotient pairing);
- mode stability, quasinormal modes, or ringdown waveforms, or the connection-Wronskian zeros that would define them: the certified complex-frequency statement is only that the axial modes and cross current continue *meromorphically* with exact singular set $\{i, i/2\}$ (polar not activated), which is not a spectrum;
- a second-order physical-process first law or nonlinear horizon dynamics;
- asymptotically flat conformal-gravity scattering with a complete boundary phase space and metric/tetrad falloffs: the axial Einstein *finite* side is now symbolic-frequency, but the divergent-class table, the polar recomputation, the regular-tetrad falloff audit and the gauge-audited phase-space/charge-algebra construction remain open;
- Hawking states, particles, grey-body factors, luminosity, a quantum BRST quotient, positivity, or unitarity: the monodromy temperature is a mode-level continuation factor, not a quantum state construction.

In particular, the paper withdraws the broad phrase “no local causal truncation.” Zero Cauchy data for the Ricci carrier is a local linear truncation. What is established is failure of the tested endpoint diagnostics to imply that truncation.

14 Certificate and reproduction ledger

The manuscript is an OUTLINE-ALLOWED/DRAFT-ALLOWED integration of committed certificates at repository baseline recorded in its generated claim map.

Claim	Authoritative certificate
Static spherical family	black_hole_programme/certificates/BH0_STATIC_SPHERICAL_BACKGROUND.json
Normalized generator and first law	black_hole_programme/certificates/BH1A_NORMALIZED_GENERATOR.json
Linear spherical dynamical extension	black_hole_programme/certificates/BH1B_DYNAMICAL_EXTENSION.json
Axial operator and Ricci–Bach composition	black_hole_programme/certificates/BH2A_AXIAL_OPERATOR.json
Extra horizon reach	black_hole_programme/certificates/BH2A_HORIZON_REACH.json
Action-derived axial current	black_hole_programme/certificates/BH2A_FLUX_MATRIX.json
Mixed and extra horizon flux fixtures	black_hole_programme/certificates/BH2A_CROSS_FLUX.json
Axial causal disposition	black_hole_programme/certificates/BH2A_CAUSAL_DISPOSITION.json
Polar trace-coupled split	black_hole_programme/certificates/BH2B_POLAR_SPLIT.json
Polar carrier horizon reach (conformal gauge quotient)	black_hole_programme/certificates/BH2B_POLAR_REACH.json
Polar Einstein two-dimensional reduction	black_hole_programme/certificates/BH2B_POLAR_EINSTEIN.json
Polar Einstein self-block null; conformal degeneracy	black_hole_programme/certificates/BH2B_POLAR_FLUX.json
Polar composition and mixed flux fixtures	black_hole_programme/certificates/BH2B_POLAR_CROSS_FLUX.json
Polar causal disposition	black_hole_programme/certificates/BH2B_POLAR_DISPOSITION.json
Horizon monodromy temperature	black_hole_programme/certificates/BH4_HAWKING_MONODROMY.json
Static-sector log classification	black_hole_programme/certificates/BH2_OMEGA_ZERO.json
Log-free carrier Jordan structure at infinity	black_hole_programme/certificates/BH2C_ASYMPOTIC_JORDAN.json
Leading metric enhancement and flux symbol	black_hole_programme/certificates/BH2C_METRIC_LEADING.json
Composed log tails; Einstein norm-selection at infinity	black_hole_programme/certificates/BH2C_FLUX_CLASS.json
Polar composed classes; parity-complete norm-selection	black_hole_programme/certificates/BH2C_POLAR_FLUX_CLASS.json
Local Cauchy truncation selects Einstein modulo conformal gauge	black_hole_programme/certificates/BH_LOCAL_EINSTEIN_CAUCHY_TRUNCATION.json
Cross invariant $a(\omega)$; axial symbolic-frequency Einstein finiteness	black_hole_programme/certificates/BH2_SYMBOLIC_CROSS_INVARIANT.json; black_hole_programme/certificates/BH2C_SYMBOLIC_FLUX_RADIATION_CLASS.json
Endpoint selection and invariant pairing; complex-frequency continuation; exterior BVP disposition	black_hole_programme/certificates/BH_ENDPOINT_NONSELECTION_ASSEMBLY.json; black_hole_programme/certificates/BH3_ANALYTIC_CONTINUATION_GATE.json; black_hole_programme/certificates/BH3_EXTERIOR_BVP_WELLPOSEDNESS_GATE.json

Every certificate records a schema, exact or controlled-numerical payload, independent verifier, mutations or controls, command receipt, dependency tags, and missing claims. This manuscript must not advance to THEOREM-FROZEN until the mixed/extra flux fixtures acquire rigorous symbolic

or interval nonvanishing bounds, the divergent-class symbolic table is completed in both parities, and the regular-tetrad falloff and asymptotic phase-space constructions are certified; the axial Einstein *finite* side is now symbolic-frequency, the asymptotic Jordan/metric analysis and the polar parity-completion gate are closed at the fixture level, and the static certificate’s complementary-sheet wording is repaired.

15 Outlook

The repeated-root asymptotic gate

first-order matrix and Jordan form $\longrightarrow$ metric reconstruction $\longrightarrow$ finite-flux boundary class

has now run past the fixture level: log-free carrier Jordan structure, log-tailed (axial) and power-enhanced single-log (polar) composed metrics, and the Einstein-only finite-slice-norm class at infinity, with the polar $\mu = 0$ Einstein pair identically zero in the slice density. Its *finite* side is now symbolic: the axial Einstein slice norm is ω -independently finite ($F^v \sim r^{-2}$) for every real $\omega \neq 0$, and the Einstein–extra pairing carries a representative-independent rank-two signature-(1, 1) invariant. The remaining completions are the divergent-class symbolic table in both parities, the regular-tetrad falloff audit, and the asymptotically flat phase-space and charge-algebra construction that would promote the Einstein finiteness to a full theorem about which flux reaches null infinity. In parallel, one rigorous interval or symbolic horizon-pairing fixture should replace the present controlled numerical nonvanishing of the extra self-pairing, whose *sign* (unlike the pairing rank and signature) is still not representative-invariant. The complex-frequency picture is now opened only as far as an exact meromorphic continuation of the axial modes and cross current, with singular set $\{i, i/2\}$ and the Einstein boundary-value problem well-posed away from the discrete zeros of its connection Wronskian; that Wronskian and its zeros — the confluent-Heun connection object — are exactly what a quasinormal computation would need, and their activation remains gated on coordinator review.

The present result already fixes an important baseline:

The axial non-Einstein curvature sector contains future-horizon-regular solutions and has nonzero controlled Lee–Wald pairing fixtures. Horizon analyticity and the presently tested leading asymptotic diagnostics do not select the Einstein kernel.

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